

10) Naive Baye's Algorithm

① Independent Events.

One event is not impacting the occurrence of another event

Rolling a Dice

{1, 2, 3, 4, 5, 6}

$$Pr(1) = \frac{1}{6}$$

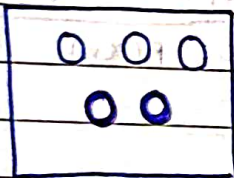
$$Pr(2) = \frac{1}{6}$$

$$Pr(3) = \frac{1}{6}$$

All are independent events they are not impacting each other.

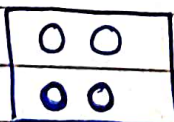
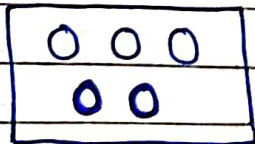
② Dependent Events

1st event we want to find the probability of finding a red marble.



$$P(R) = \frac{3}{5}$$

2nd event we want to find the probability of finding a green marble given that the first marble is red.



$$P(G) = \frac{2}{4}$$

This is dependent event

$$P(B/A) = \frac{P(B) * P(A/B)}{P(A)}$$

Bayes Theorem

$x_1 \quad x_2 \quad x_3 \quad y$

dependent variable

$$P(y/x_1, x_2, x_3) = \frac{P(y) * P(x_1, x_2, x_3/y)}{P(x_1, x_2, x_3)}$$

Independent variables

$$= \frac{P(y) * P(x_1/y) * P(x_2/y) * P(x_3/y) * \dots * P(x_n/y)}{P(x_1) * P(x_2) * P(x_3) * \dots * P(x_n)}$$

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